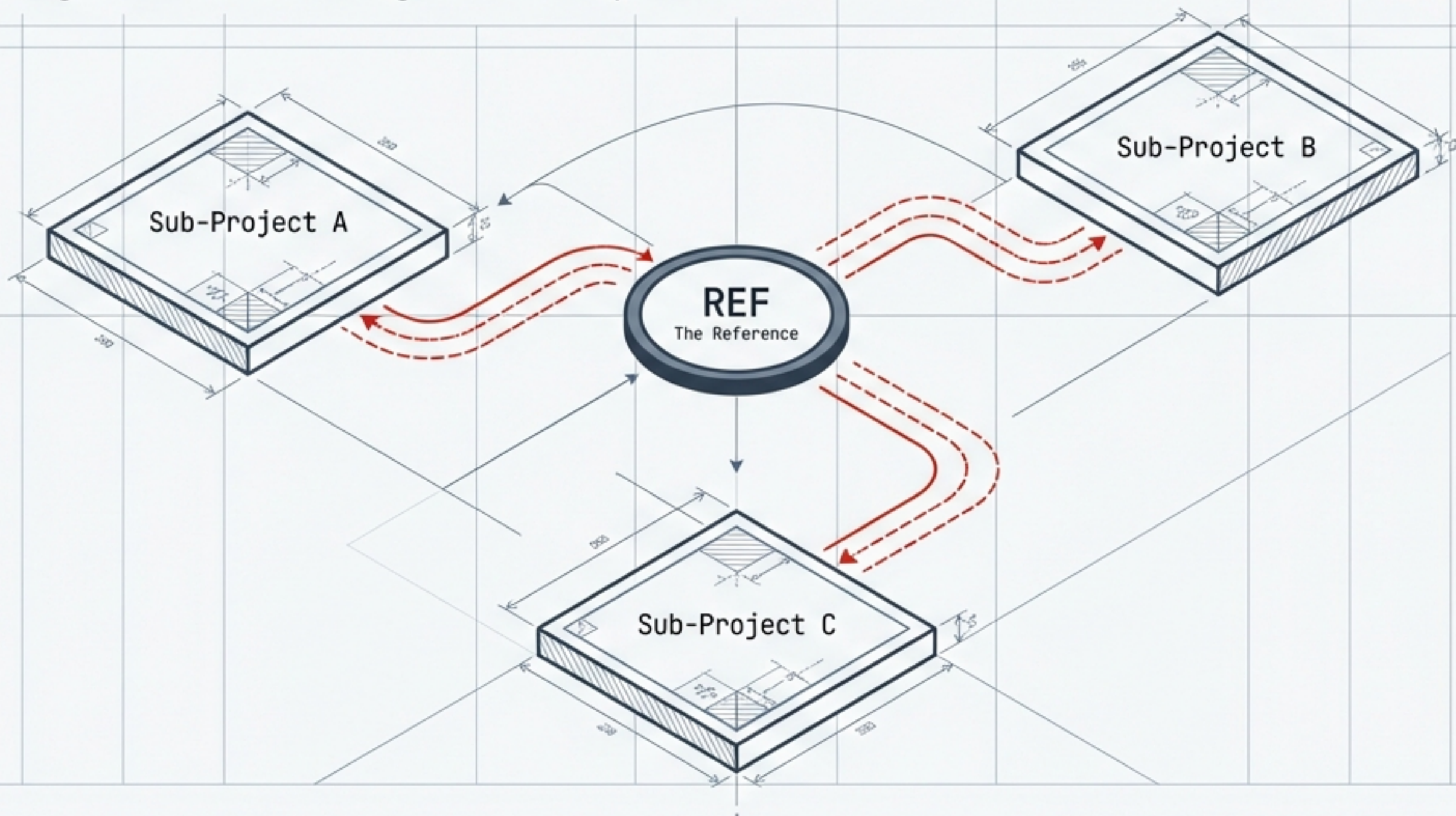
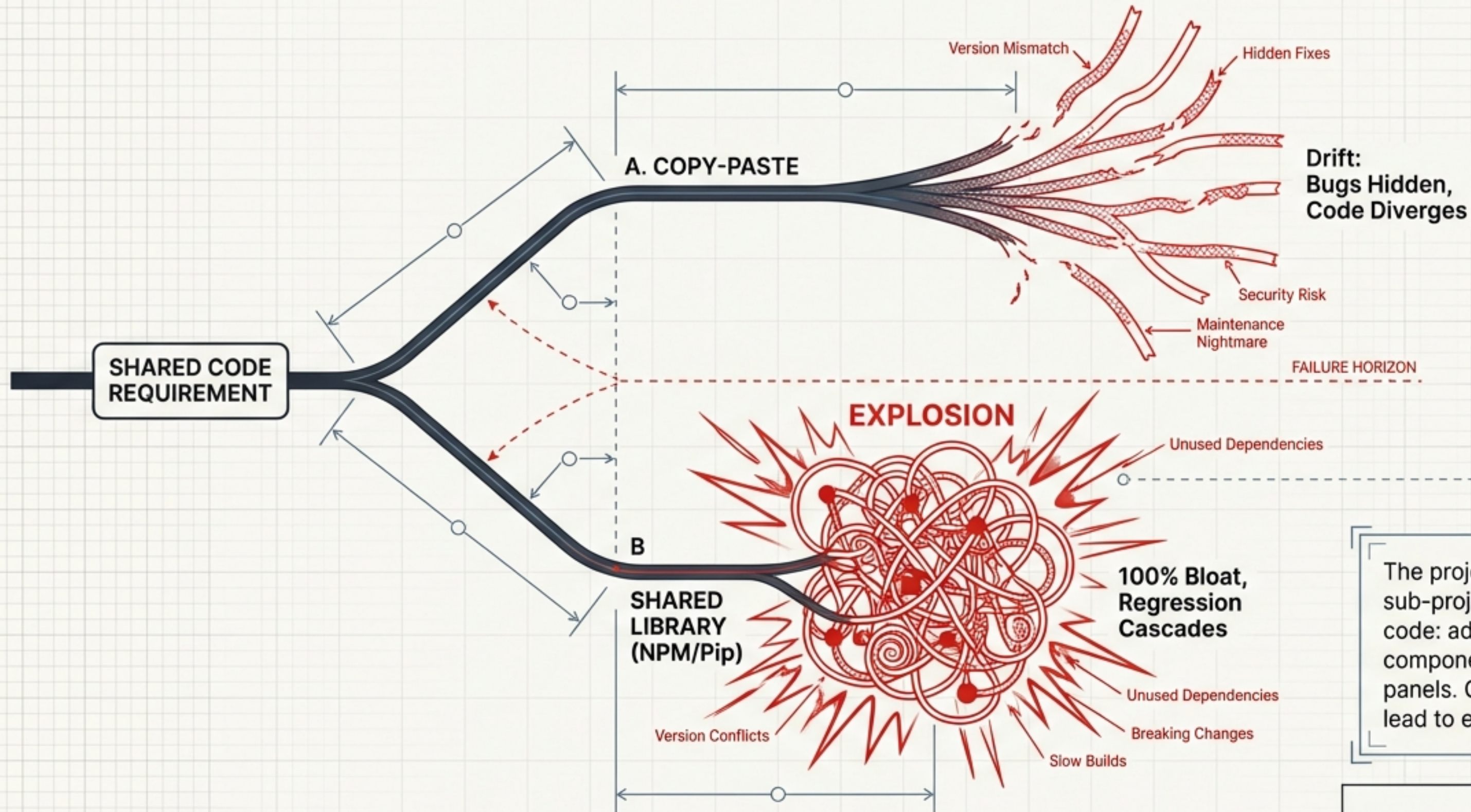


Architecture Brief: Reference Implementations

A Dependency Management Pattern for Agentic Development

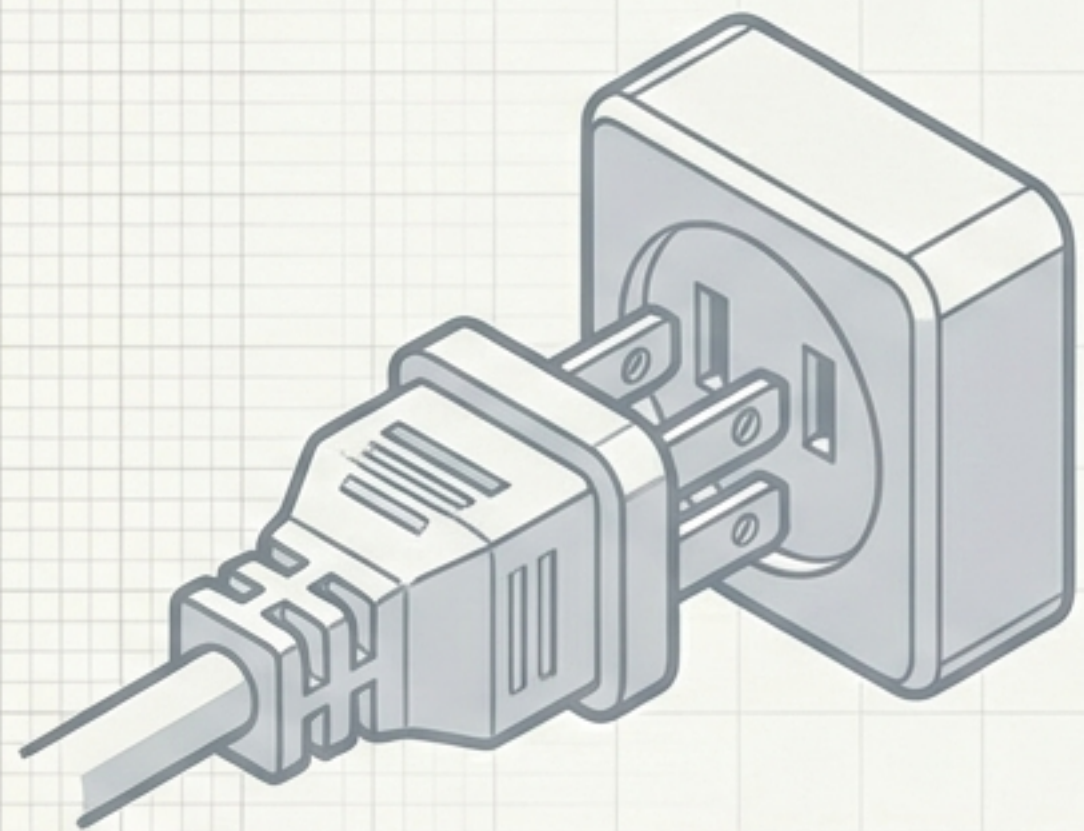


The Dependency Dilemma in Shared Code

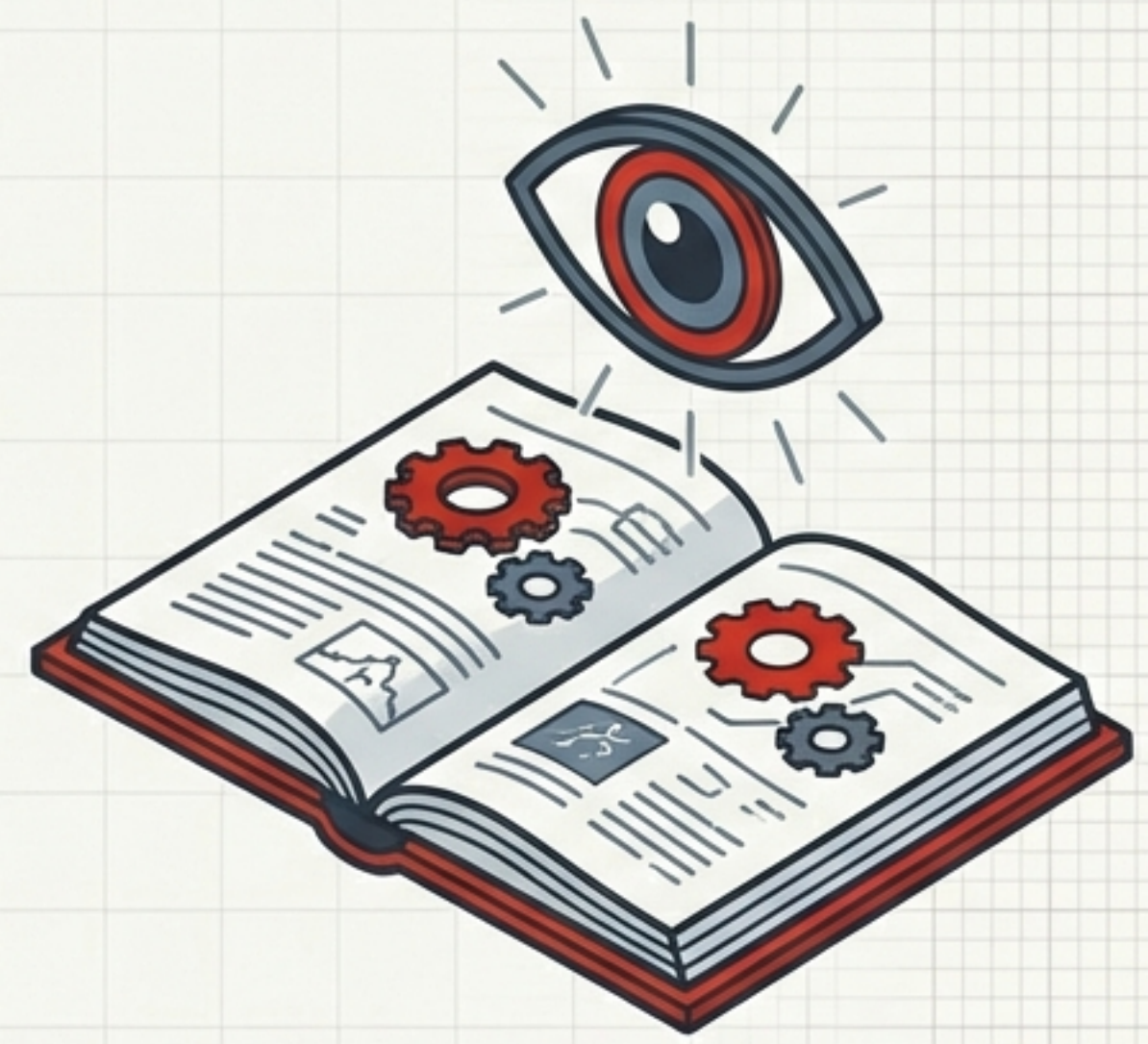


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The Third Option: Reference Implementations



Runtime Import
Rigid Dependency



LLM Consumption
Agentic Adaptation

A standalone repo containing the canonical, well-documented version of a shared capability. Designed to be read by LLMs, not imported as a dependency.

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Shifting the Paradigm

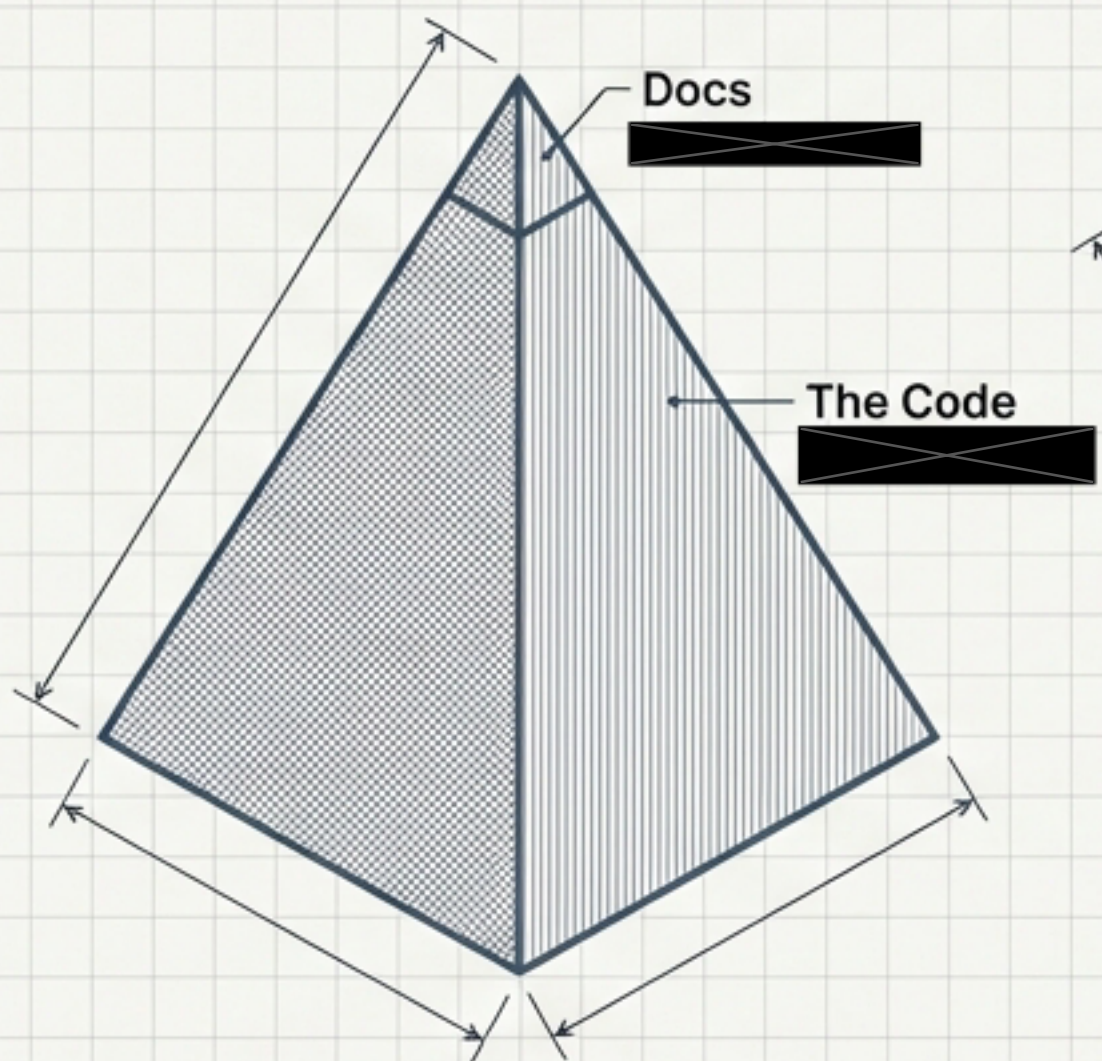
Aspect	Traditional Library	Reference Implementation
Integration	<code>npm install</code> (Blind Import)	LLM Reads & Adapts
Updates	Automatic Breaking Changes	LLM Reviews Fix per Project
Payload	100% Code Load	2-10% Code Load (Need only)
Conflict	Dependency Hell	Independent Adaptation

"Bug fix in reference → LLM reviews the fix and adapts it.
No automatic breaking changes."

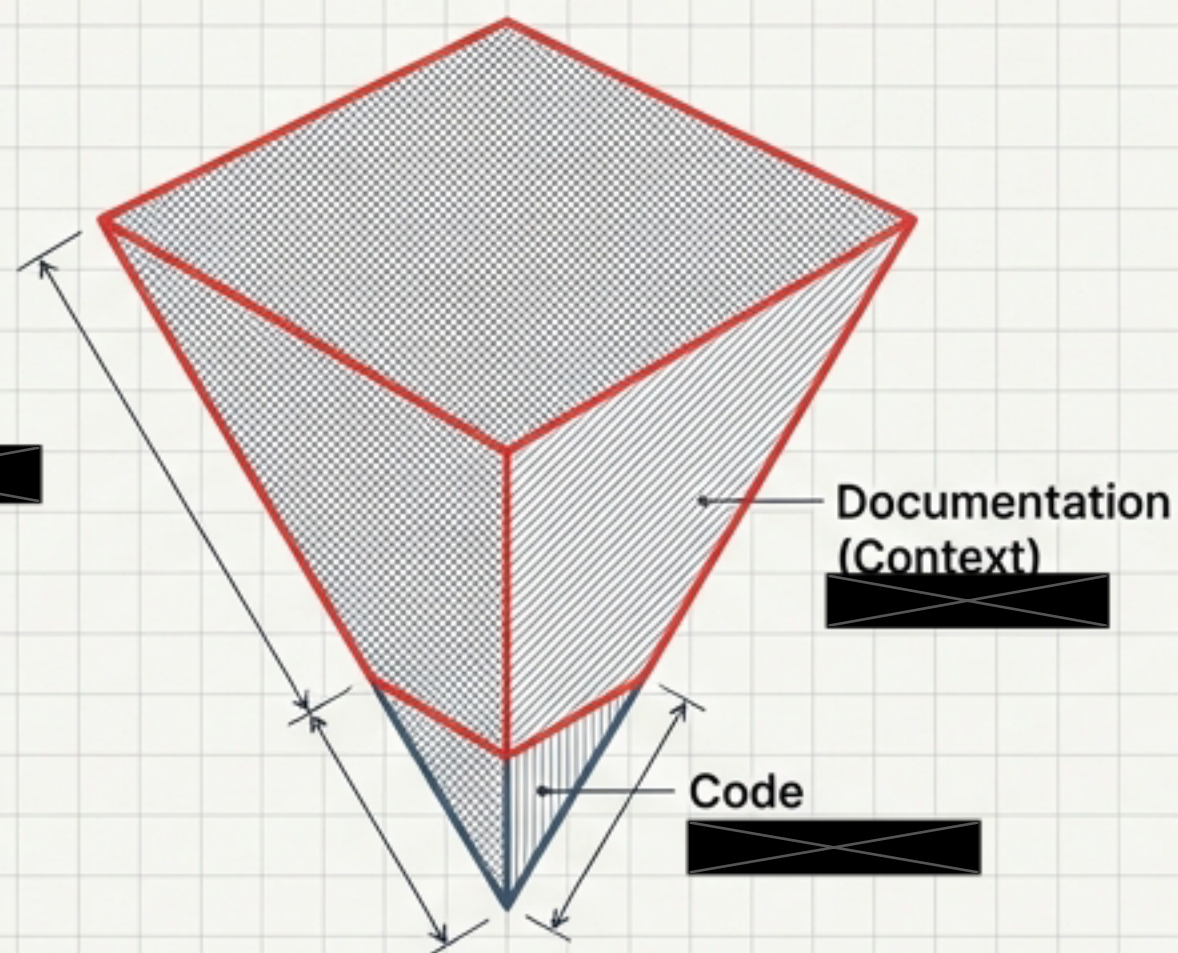
PROJECT:	
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Documentation is the Interface

Traditional Library



Reference Pattern



The reference repo is over-indexed on documentation for LLM comprehension.

The code is good, but the documentation is what makes the pattern work.

The docs serve as the API for the Agent's understanding.

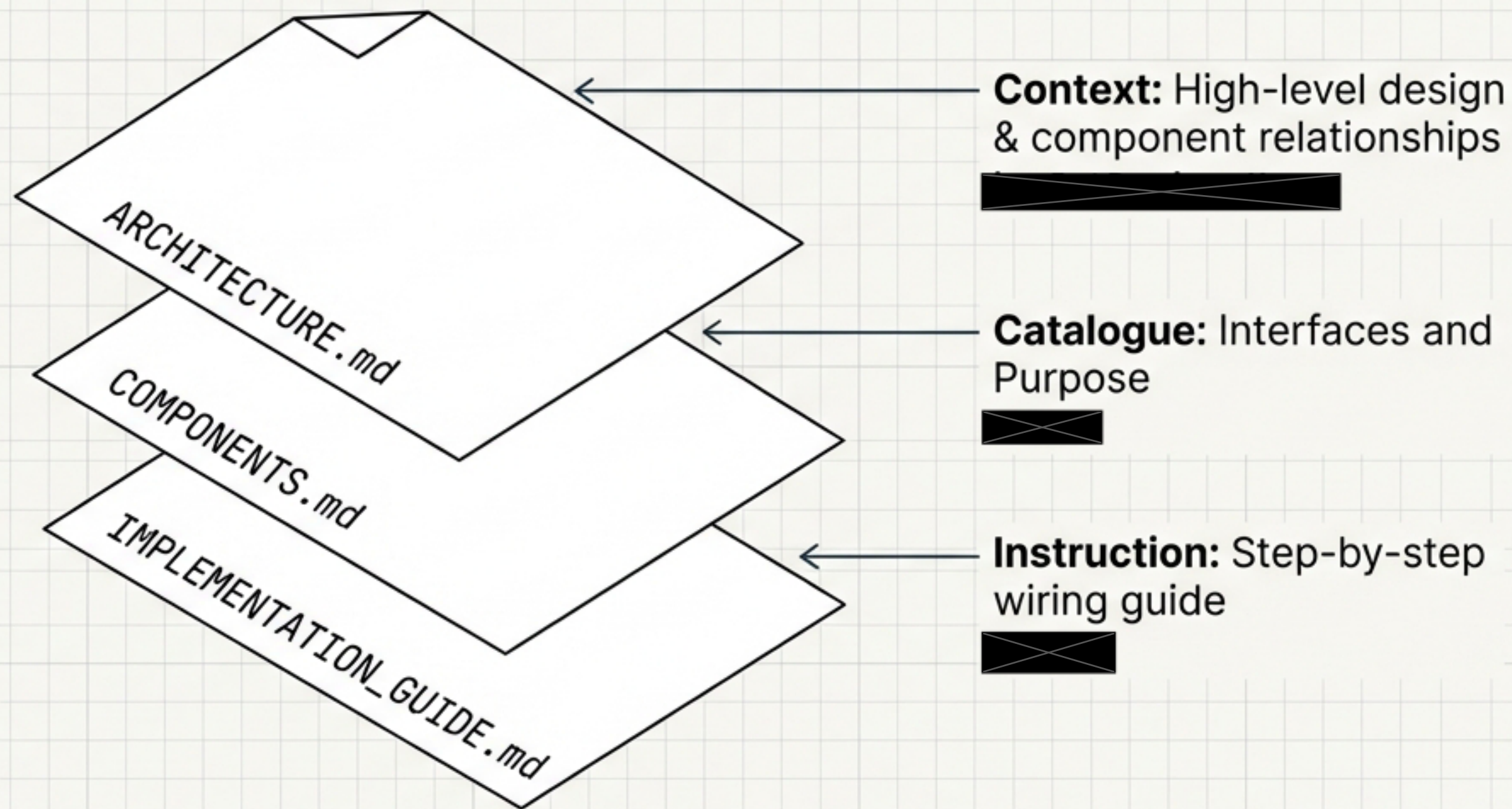
PROJECT:
REFERENCE IMPLEMENTATION STUDY

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A-003

DATE:
OCT 26, 2023

SCALE:
NTS

The LLM Consumption Suite

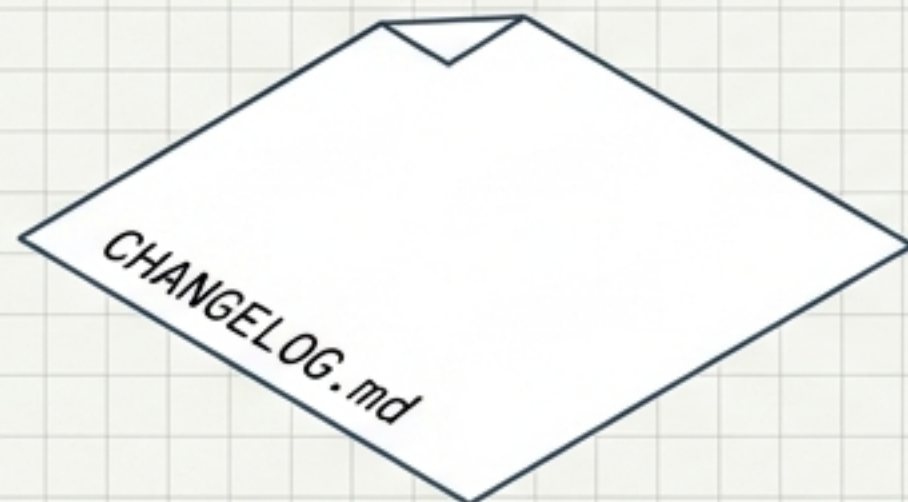


These files allow the LLM to understand **what** to take and **how** to wire it up, rather than just copying bytes.

PROJECT: REFERENCE IMPLEMENTATION STUDY		
DRAWING NO: A-004	DATE: NOV 15, 2023	SCALE: NTS

Managing Change and Verification

The Context



Detailed 'Why' and 'How'.
Not just dates.

Example Text:

v1.10: Added user
confirmation dialogs.

The Verification

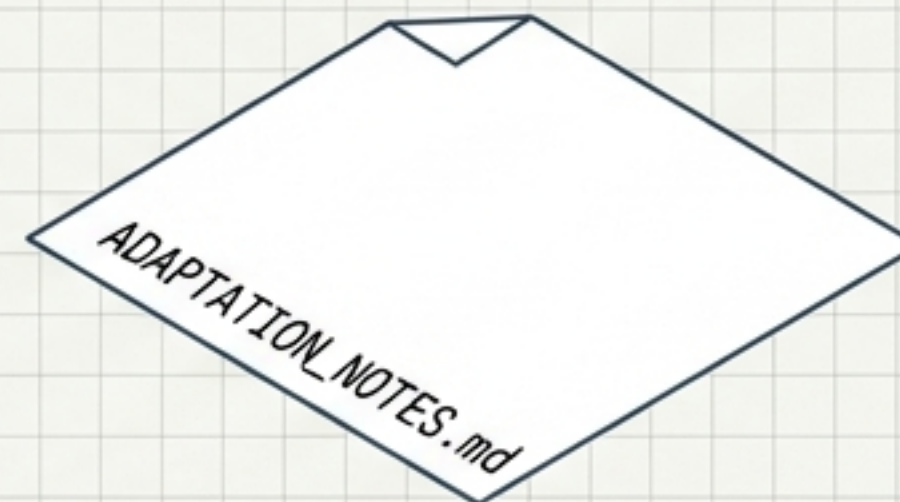


Expected inputs & outputs.

Example Text:

Dispatch sg-message,
verify dialog renders.

The Nuance



Known variations.

Example Text:

SG/Send uses dark theme
—override --msg-bg.

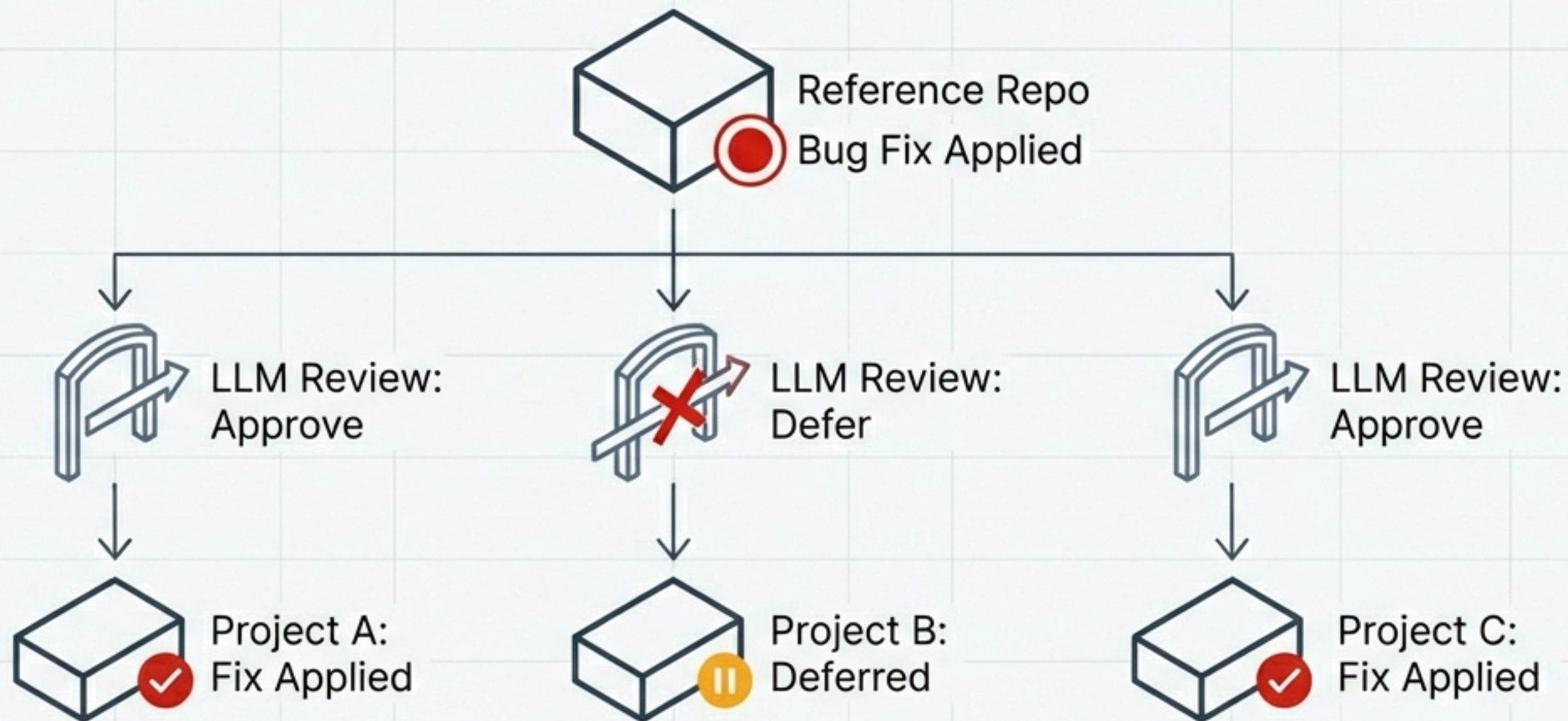
PROJECT: REFERENCE IMPLEMENTATION STUDY		
DRAWING NO: A-005	DATE: NOV 22, 2023	SCALE: NTS

Workflow: Adding a New Feature



PROJECT: REFERENCE IMPLEMENTATION STUDY		
DRAWING NO: A-006	DATE: NOV 29, 2023	SCALE: NTS

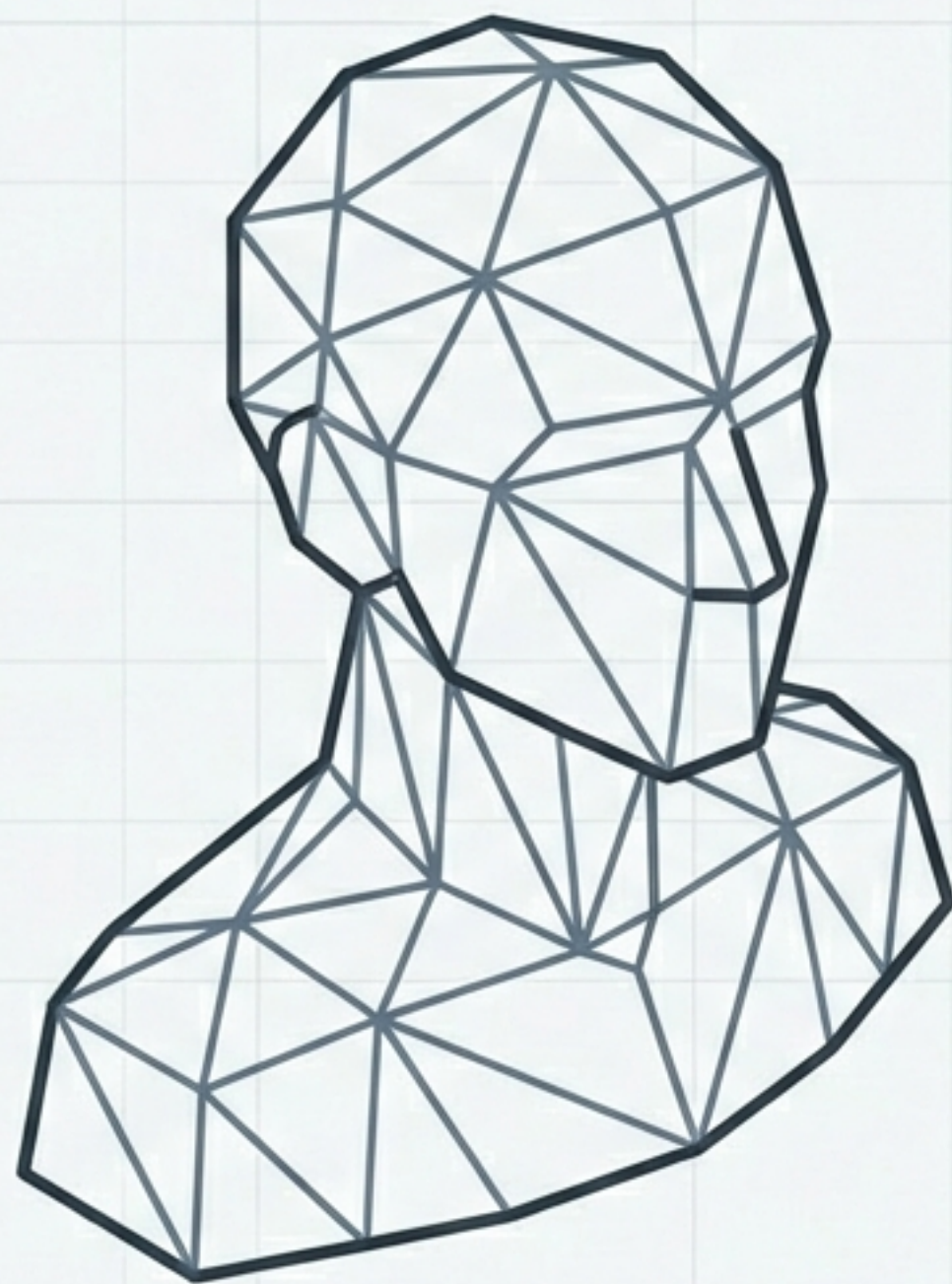
Workflow: Propagating a Bug Fix



No "Break one, Break all". Sub-projects choose when to sync based on LLM review.

PROJECT: REFERENCE IMPLEMENTATION STUDY		
DRAWING NO: A-007	DATE: NOV 30, 2023	SCALE: NTS

The Reference Agent Persona



Capabilities

- Explains design decisions (Why did we do it this way?)
- Answers implementation questions via chat
- Guides the Sub-project LLM during adaptation
- Reviews adaptations and flags deviations

Loaded via `context.md`

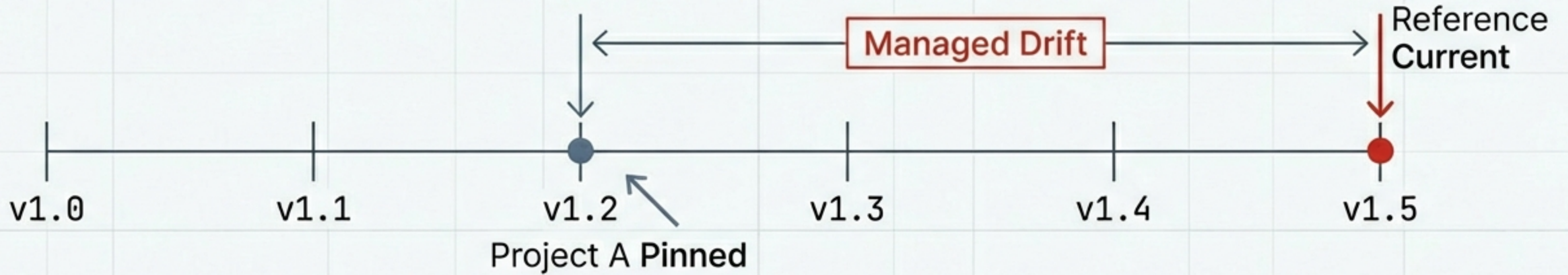
PROJECT:
REFERENCE IMPLEMENTATION STUDY

DRAWING NO:
A-008

DATE:
NOV 30, 2023

SCALE:
NTS

Controlled Drift & Version Pinning

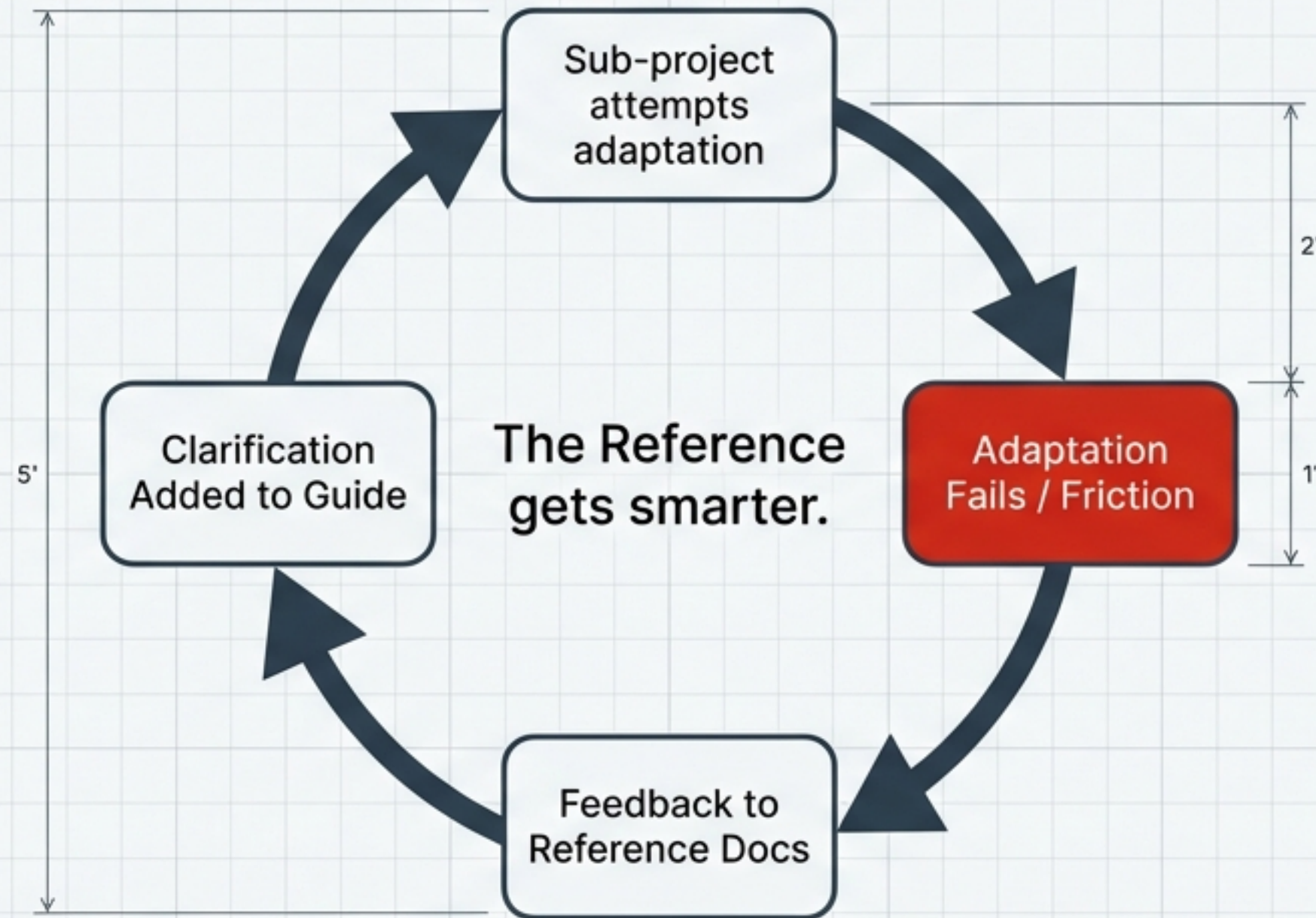


- > ANALYSIS: You are 3 versions behind.
- > MISSING: 'API Debug Panel' feature.
- > CUSTOM: CSS --msg-bg is overridden.

Drift is no longer accidental; it is a visible, managed decision.

PROJECT: REFERENCE IMPLEMENTATION STUDY		
DRAWING NO: A-009	DATE: NOV 30, 2023	SCALE: NTS

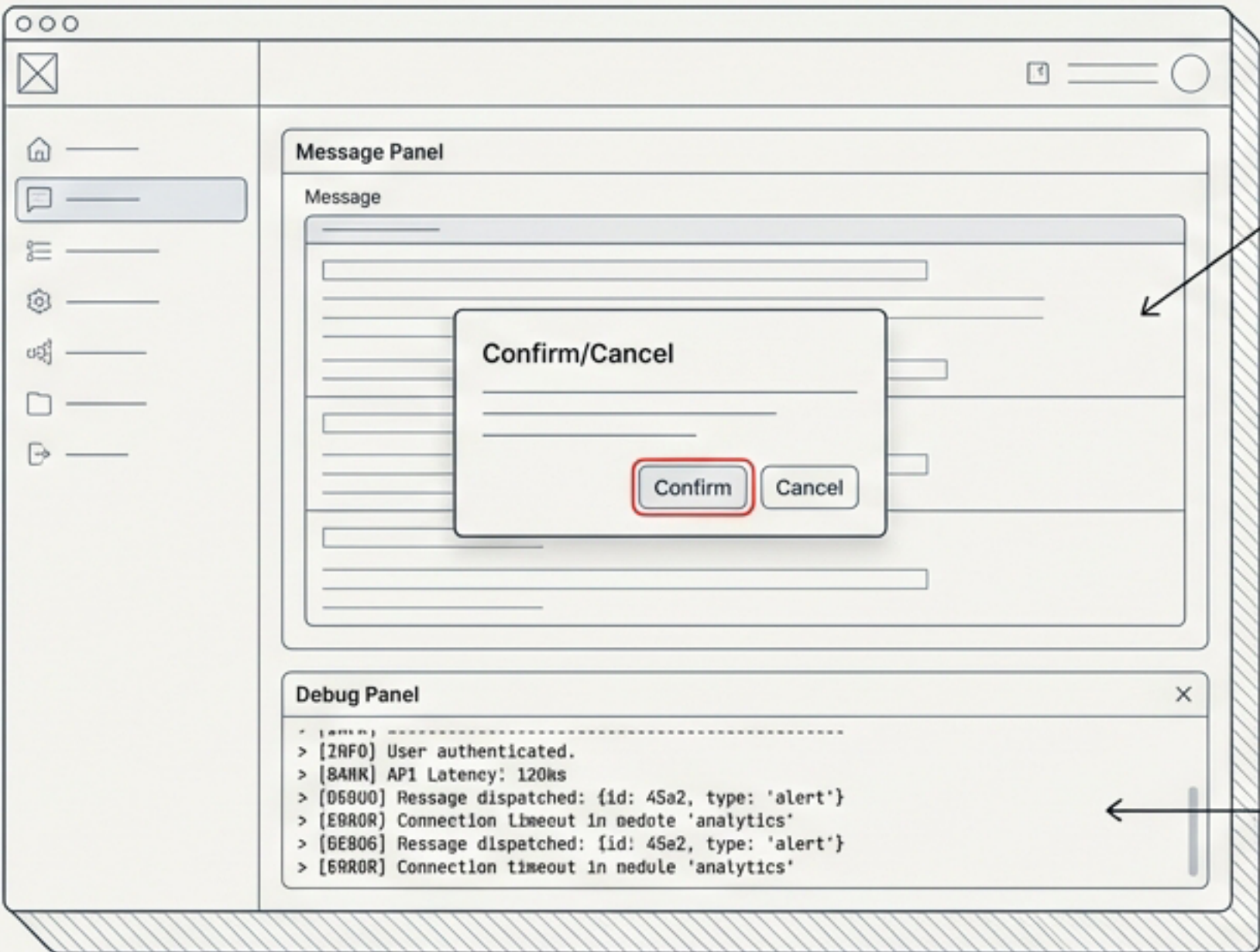
The Self-Correction Loop



Every failed adaptation makes the reference better.
The documentation evolves through usage.

project:	drawing no:
REFERENCE IMPLEMENTATION STUDY	A-010
date:	scale:
NOV 30, 2023	NTS

Pilot: The Admin UI Pattern



Ref-Component:
Messaging

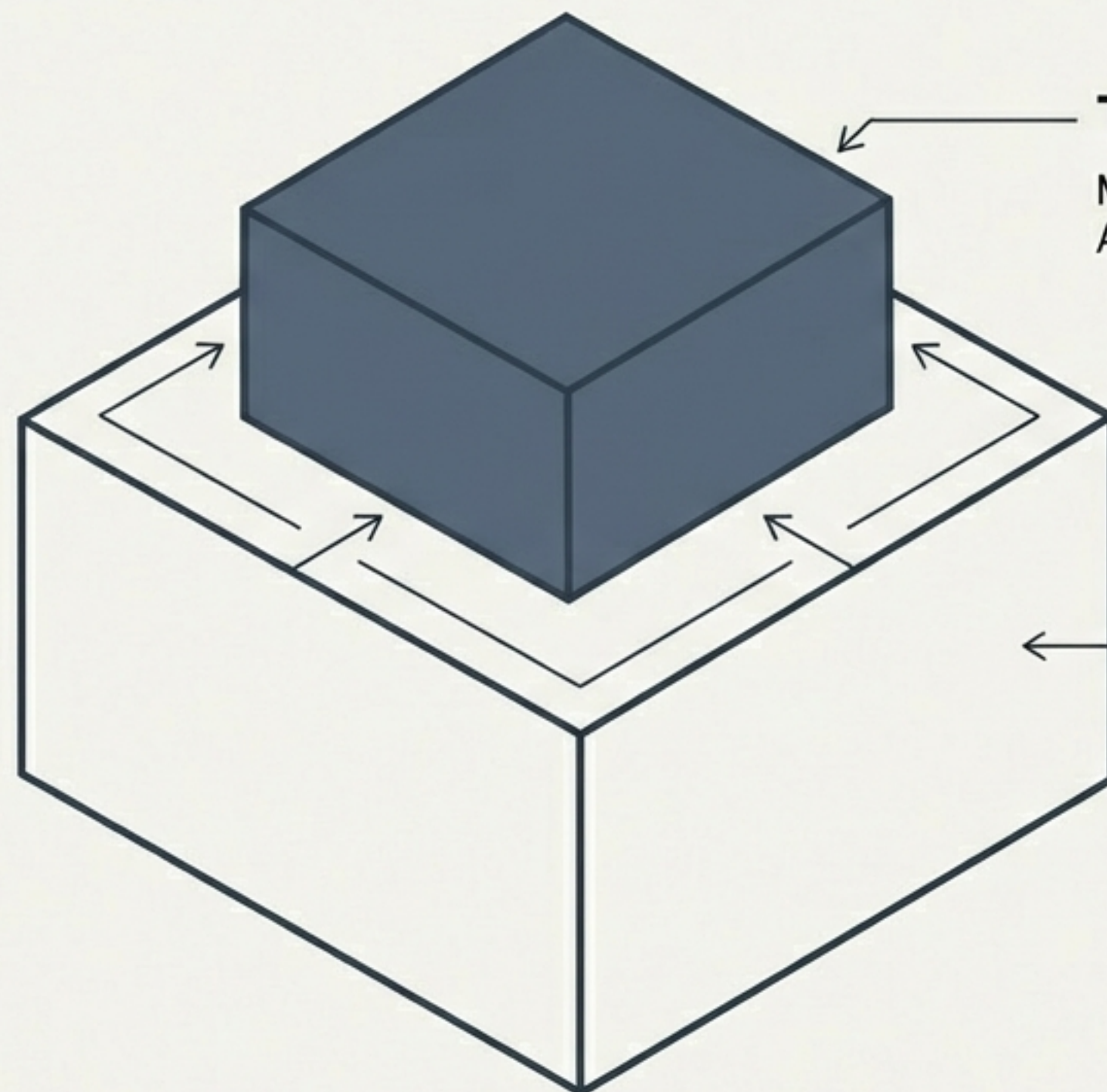
Ref-Component:
Debugger

Inter	JetBrains Mono
Repo	ref-admin-ui
Features	Event-driven architecture, Shared message bus
Consumers	• SG/Send Admin • QA Runner • +2 others

Pilot architecture establishes the standardized admin experience.
Consistency through shared patterns.

PROJECT: REFERENCE IMPLEMENTATION STUDY		
DRAWING NO: A-011	DATE: NOV 30, 2023	SCALE: NTS

A New Commercial Model for Open Source



The Intelligence
Maintenance. Guidance.
Agent Persona. Billable.

The Code
Open Source. Free.
Replicated.

- Value moves from “The Code” to “The Guidance”.
- Billable service: Ongoing curation & adaptation support.
- The Agent is the product.

PROJECT: REFERENCE IMPLEMENTATION STUDY		
DRAWING NO: A-012	DATE: DEC 05, 2023	SCALE: NTS
DRAWN BY: AI-ARCHITECT		

Definition of Success

- ☐ Reference `ref-admin-ui` repo created.
- ☐ `ARCHITECTURE.md` and `COMPONENTS.md` complete.
- ☐ `IMPLEMENTATION_GUIDE.md` ready for LLM.
- ☐ At least one sub-project pinned and synced.
- ☐ Reference Agent persona active.

The code is free. The intelligence is the product.

PROJECT: REFERENCE IMPLEMENTATION STUDY		
DRAWING NO: A-013	DATE: DEC 12, 2023	SCALE: NTS

DRAWN BY: AI-ARCHITECT